

Memory units, RAM, ROM and measurement ★ Golden

One college period · 40–45 minutes · Computer Memory

What you will learn

- State bit, nibble, byte and the binary prefixes KB–TB
- Differentiate volatile and non-volatile memory
- Explain RAM (SRAM, DRAM) and ROM (PROM, EPROM, EEPROM)
- Compare primary and secondary memory

Unit	Size
Bit	0 or 1 — smallest unit of data
Nibble	4 bits
Byte	8 bits — stores one character
KB (Kilobyte)	1024 bytes
MB (Megabyte)	1024 KB
GB (Gigabyte)	1024 MB
TB (Terabyte)	1024 GB

RAM

volatile · read/write
holds running programs
SRAM / DRAM

ROM

non-volatile · mostly read
holds BIOS / firmware
PROM / EPROM / EEPROM

Primary memory: RAM vs ROM.

LEARN IT

Primary / main memory sits on the motherboard and is directly used by the CPU. RAM is volatile: its contents vanish when power is cut. ROM is non-volatile: it keeps BIOS even when the PC is off.

RAM	ROM
Random Access Memory	Read Only Memory
Read and write	Mostly read; writing needs special process
Volatile	Non-volatile
Stores OS + running programs + data	Stores firmware / BIOS
Larger, cheaper per bit (DRAM)	Smaller

RAM types

- SRAM (Static): uses flip-flops, faster, used as cache, more expensive, no refresh.
- DRAM (Dynamic): uses capacitors, needs refresh, cheaper, used as main memory.

- SIMM = Single Inline Memory Module (board MCQ).

ROM types

- PROM: programmed once by the user.
- EPROM: erased with ultraviolet light, then reprogrammed.
- EEPROM: electrically erased and reprogrammed, used for modern BIOS.

HOW BIEK ASKS THIS

MCQ: 1 GB = 1024 MB. Never write 1000. Difference of primary vs secondary: speed, volatility, capacity, CPU distance, cost.

COMMON BOARD MISTAKE

1 GB is 1024 MB, not 1024 KB. Cache is high-speed SRAM inside or near the CPU, not the hard disk.

اردو میں سمجھیں

سب سے چھوٹی اکائی بٹ ہے (0 یا 1)۔ آٹھ بٹ کا بائٹ ایک حرف رکھتا ہے۔ 1024 بائٹ = 1 KB۔
RAM عارضی ہے، بجلی جانے پر خالی۔ ROM مستقل ہے، BIOS رکھتی ہے۔ SRAM تیز کیش ہے، DRAM مین میموری ہے۔

CHECK YOURSELF

1. Convert: 2 KB = ? bytes
2. Which ROM is erased by UV light?

Answers: 1. $2 \times 1024 = 2048$ bytes · 2. EPROM

MCQs

1. 1 Gigabyte is equivalent to:
 - a. 1024 KB
 - b. 1024 Bytes
 - c. 1024 MB
 - d. 1024 TB
2. SIMM is the abbreviation of:
 - a. Single interfacing Memory Module
 - b. Simple Inline Memory Module
 - c. Single Inline Memory Module
 - d. Simple Inline Memory Mode
3. Smallest unit of data is:
 - a. Bit
 - b. Byte
 - c. Megabyte
 - d. Nibble

Key: 1=c, 2=c, 3=a

Short questions (3 marks each)

1. Differentiate RAM and ROM.
2. Explain SRAM and DRAM.

3. Write memory measurement units from bit to TB.

Model short answers

1. RAM: volatile R/W working memory. ROM: non-volatile firmware memory. RAM loses data without power; ROM does not.
2. SRAM is fast cache without refresh. DRAM is cheaper main memory that must be refreshed.
3. Bit, nibble (4 bits), byte (8 bits), KB, MB, GB, TB — each step $\times 1024$ except bit/nibble/byte.

Long questions (board Section C)

1. What is computer memory? Explain primary memory and its types with a comparison table.